Improving GARCH volatility forecasts with regime-switching GARCH

Abstract and Introduction

This paper addresses a significant issue with standard Generalized Autoregressive Conditional Heteroskedasticity (GARCH) models: their tendency to produce volatility forecasts that are too high in volatile periods. This problem is attributed to the high persistence of shocks inherent in GARCH forecasts, which may stem from unmodeled structural changes in the variance process. To enhance flexibility regarding volatility persistence, the paper introduces a Markov regime-switching GARCH (RS-GARCH) model, which distinguishes between two volatility regimes and allows for GARCH effects within each regime. This model resolves the issue of excessive forecasts and yields significantly better out-of-sample volatility forecasts for major U.S. dollar exchange rates.

Good volatility forecasts are crucial for financial decisions, such as currency option pricing. While GARCH models effectively capture volatility clustering, they struggle with adequately describing volatility persistence when structural changes occur in the variance process. A key contribution of this paper's RS-GARCH model is its ability to perform multi-period-ahead volatility forecasting through a convenient recursive procedure, distinguishing it from existing variants like Gray's (1996a) specification.

2. Regime-switching GARCH Model

2.1. The Model Specification

The RS-GARCH model comprises four components: the mean, regime process, variance, and error distribution.

Mean Equation Let S_t be the logarithm of a spot exchange rate at time t. The exchange rate change, s_t, is defined as the percentage depreciation of the domestic currency:

$$s_t = 100(S_t - S_{t-1}) \tag{1}$$

The mean equation for s_t is modeled as a random walk with drift:

$$s_t = \mu + e_t \tag{2}$$

Here, μ is the constant conditional mean of s_t, and e_t is an innovation with zero mean, conditional on the information set.

Regime Process The unobserved variance regime at time t is denoted by $r_t \in \{1, 2\}$, where $r_t = 1$ signifies the low-variance regime. The regimes are assumed to follow a first-order Markov process with constant staying probabilities:

$$p_{t-1}(r_t|\tilde{r}_{t-1}) = p(r_t|r_{t-1}) = \begin{cases} p_{11} & \text{if } r_t = r_{t-1} = 1 \\ p_{22} & \text{if } r_t = r_{t-1} = 2 \end{cases} \tag{3}$$

High values of p_{11} and p_{22} ensure the required regime persistence, allowing the model to explain part of the overall volatility persistence. The full information set at time $t-1$ includes observable information $I_{t-1} = (s_{t-1}, s_{t-2}, \ldots)$ and the unobserved regime path $\tilde{r}_{t-1} = (r_{t-1}, r_{t-2}, \ldots)$.

Conditional Variance A direct application of the GARCH(1,1) model in a regime-switching context would be:

$$V_{t-1}(e_t|\tilde{r}_t) = \omega_{r_t} + \alpha_{r_t} e_{t-1}^2 + \beta_{r_t} V_{t-2}(e_{t-1}|\tilde{r}_{t-1}) \tag{4}$$

This specification is deemed practically infeasible for estimation due to its dependence on the entire unobserved regime path $\tilde{r}_t$, leading to an exponentially growing number of paths.

To circumvent this "path dependence" problem, previous models like Cai (1994) and Hamilton and Susmel (1994) removed the GARCH term, relying only on ARCH effects. Gray (1996a) proposed integrating out the unobserved regime path in the GARCH term itself:

$$V_{t-1}(e_t|\tilde{r}_t) = \omega_{r_t} + \alpha_{r_t} e_{t-1}^2 + \beta_{r_t} E_{t-2}[V_{t-2}(e_{t-1}|\tilde{r}_{t-1})] \tag{5}$$

While this retains GARCH effects and speeds up estimation, multi-period-ahead forecasting remains complicated.

The proposed regime-switching GARCH(1,1) model in this paper modifies Gray's approach by postponing the integration of the previous regime until time $t-1$ and leveraging information from the current regime r_t:

$$\mathbf{V_{t-1}}(\mathbf{e_t}|\tilde{\mathbf{r}}_\mathbf{t}) = \omega_{\mathbf{r_t}} + \alpha_{\mathbf{r_t}}\mathbf{e_{t-1}^2} + \beta_{\mathbf{r_t}}\mathbf{E_{t-1}}[\mathbf{V_{t-2}}(\mathbf{e_{t-1}}|\tilde{\mathbf{r}}_\mathbf{t-1})|\mathbf{r_t}] \tag{6}$$

The expectation on the right-hand side is across the regime path $\tilde{r}_{t-1}$, conditional on the observable information I_{t-1} and the current regime r_t. This construction ensures that $V_{t-1}(e_t|\tilde{r}_t)$ depends only on r_t, i.e., $V_{t-1}(e_t|\tilde{r}_t) = V_{t-1}(e_t|r_t)$, thereby solving the path dependence issue. For the conditional variance to be strictly positive, the parameters must satisfy: $\omega_{r_t} > 0$, $\alpha_{r_t} \geq 0$, and $\beta_{r_t} \geq 0$ for all regimes r_t.

Conditional Distribution The innovation e_t, conditional on I_{t-1} and $\tilde{r}_t$, is assumed to follow a t-distribution:

$$e_t|I_{t-1}, \tilde{r}_t \sim t(\nu, 0, V_{t-1}(e_t|r_t)) \tag{7}$$

where ν represents the degrees of freedom (independent of conditioning information), and $V_{t-1}(e_t|r_t)$ is the conditional variance. The use of a t-distribution, which includes the normal distribution as a limiting case, enhances the stability of the regimes by better accommodating large outliers without triggering premature regime switches.

In essence, the complete regime-switching GARCH model is defined by equations (1), (2), (5), and (6). It generalizes the standard one-regime GARCH(1,1) model, which can be seen as a special case where all regime-specific parameters are identical.

2.2. Properties of the Model

2.2.1. Flexibility Regarding Volatility Persistence The RS-GARCH model significantly increases flexibility in capturing volatility persistence. Shocks can now be followed by volatile periods not only due to GARCH effects but also due to a switch to a high-volatility regime. Furthermore, allowing for distinct ARCH and GARCH parameters across regimes enables the model to capture more complex dynamics, such as greater shock persistence in high-volatility periods or "pressure relieving" effects where large shocks are followed by tranquil periods.

2.2.2. Recursive Volatility Forecasting For forecasting the variance of the h-period exchange rate change, $s_{t,h} = \sum_{j=0}^{h-1} s_{t+j}$, conditional on information at $t-1$, $V_{t-1}(s_{t,h})$: Due to the assumed absence of serial correlation in one-period changes, the h-period variance can be expressed as the sum of conditional one-period variances:

$$V_{t-1}(s_{t,h}) = \sum_{i=0}^{h-1} V_{t-1}(s_{t+i}) \tag{8}$$

Each one-period variance is computed as a weighted sum over the possible regimes:

$$V_{t-1}(s_{t+i}) = \sum_{r_{t+i}=1,2} p_{t-1}(r_{t+i}) \cdot V_{t-1}(e_{t+i}|r_{t+i}) \tag{9}$$

where $p_{t-1}(r_{t+i})$ is the probability of being in regime r_{t+i} at time $t+i$, conditional on I_{t-1}. A core advantage of the proposed model is that $V_{t-1}(e_{t+i}|r_{t+i})$ can be computed recursively by iterating forward on:

$$\mathbf{V_{t-1}}(\mathbf{e_{t+i}}|\mathbf{r_{t+i}}) = \omega_{\mathbf{r_{t+i}}} + (\alpha_{\mathbf{r_{t+i}}} + \beta_{\mathbf{r_{t+i}}}) \cdot \mathbf{E_{t-1}}[\mathbf{V_{t-1}}(\mathbf{e_{t+i-1}}|\mathbf{r_{t+i-1}})|\mathbf{r_{t+i}}] \tag{10}$$

The term $p_{t-1}(r_{t+i})$ in (8) is calculated using the time-constant Markov transition matrix M:

$$M = \begin{pmatrix} p_{11} & 1-p_{22} \\ 1-p_{11} & p_{22} \end{pmatrix} \tag{11}$$

The multi-period-ahead regime probability is then $p_{t-1}(r_t|r_{t-1}) = (M^{t-(t-1)})_{r_t r_{t-1}}$. The probability $p_{t-1}(r_t)$ is given by:

$$p_{t-1}(r_t) = \sum_{r_{t-1}=1,2} p_{t-1}(r_{t-1}) \cdot p_{t-1}(r_t|r_{t-1}) \tag{12}$$

The expectation in (9) further relies on the probability $p_{t-1}(r_{t-1}|r_t)$:

$$p_{t-1}(r_{t-1}|r_t) = \frac{p_{t-1}(r_t|r_{t-1}) \cdot p_{t-1}(r_{t-1})}{p_{t-1}(r_t)} \tag{13}$$

2.2.3. Unconditional Variance The "unconditional" error variance for each regime, $V(e_t|r_t = i) = s_i^2$, assuming it exists and is independent of t, can be derived as:

$$\begin{pmatrix} s_1^2 \\ s_2^2 \end{pmatrix} = (I_2 - A)^{-1} \begin{pmatrix} \omega_1 \\ \omega_2 \end{pmatrix} \tag{14}$$

where I_2 is the 2×2 identity matrix, and the (i, j)-th element of matrix A is defined as $A_{ij} = P(r_{t-1} = j|r_t = i)(\alpha_i + \beta_i)$. Necessary conditions for the existence of both unconditional variances are $A_{11} < 1$, $A_{22} < 1$, and $\det(I_2 - A) > 0$. The probabilities $p(r_{t-1}|r_t)$ required for A_{ij} are given by Bayes' rule:

$$p(r_{t-1}|r_t) = \frac{p(r_t|r_{t-1}) \cdot p(r_{t-1})}{\sum_{r'_{t-1}=1,2} p(r_t|r'_{t-1}) \cdot p(r'_{t-1})} \tag{15}$$

The unconditional probabilities of being in each regime, $p(r_{t-1} = 1)$ and $p(r_{t-1} = 2)$, are derived from the Markov process:

$$p(r_{t-1} = 1) = \frac{1 - p_{22}}{2 - p_{11} - p_{22}} \tag{16}$$

$$p(r_{t-1} = 2) = \frac{1 - p_{11}}{2 - p_{11} - p_{22}} \tag{17}$$

2.2.4. Recursive Estimation The RS-GARCH model is estimated using maximum likelihood (ML). The likelihood function possesses a first-order recursive structure, speeding up the estimation process. The density of s_t conditional on observable information $p_{t-1}(s_t)$ is a mixture of t-densities:

$$p_{t-1}(s_t) = \sum_{r_t=1,2} p_{t-1}(s_t|r_t) \cdot p_{t-1}(r_t) \tag{18}$$

The term $p_{t-1}(s_t|r_t)$ is the t-density derived from equations (1), (5), and (6). Its computation requires the probability $p_{t-1}(r_{t-1}|r_t)$:

$$p_{t-1}(r_{t-1}|r_t) = \frac{p_{t-1}(r_{t-1}) \cdot p_{t-1}(r_t|r_{t-1})}{p_{t-1}(r_t)} \tag{19}$$

The probability $p_{t-1}(r_t)$ is calculated as:

$$p_{t-1}(r_t) = \sum_{r_{t-1}=1,2} p_{t-1}(r_{t-1}) \cdot p_{t-1}(r_t|r_{t-1}) \tag{20}$$

The crucial component, $p_{t-1}(r_{t-1})$, also exhibits a first-order recursive structure:

$$p_{t-1}(r_{t-1}) = p_{t-2}(r_{t-1}|s_{t-1}) = \frac{p_{t-2}(s_{t-1}|r_{t-1}) \cdot p_{t-2}(r_{t-1})}{\sum_{r'_{t-1}=1,2} p_{t-2}(s_{t-1}|r'_{t-1}) \cdot p_{t-2}(r'_{t-1})} \tag{21}$$

The overall sample log-likelihood is then $\sum_{t=1}^{T} \log(p_{t-1}(s_t))$.

2.2.5. Recursive Regime Inference The model allows for the estimation of two types of regime probabilities:

- Ex ante probability ($p_{t-1}(r_t)$): The probability of being in regime r_t at time t using information I_{t-1}.
- Smoothed regime probability ($p_T(r_t)$): The probability of being in regime r_t at time t using the complete data set I_T.

More generally, any ex post regime probability $p_t(r_t)$ (for $t' \geq t$) can be computed recursively starting from the ex ante probabilities:

$$p_t(r_t) = p_{t-1}(r_t|s_t) = \frac{p_{t-1}(s_t|r_t) \cdot p_{t-1}(r_t)}{\sum_{r'_t=1,2} p_{t-1}(s_t|r'_t) \cdot p_{t-1}(r'_t)} \tag{22}$$

For $t' > t$, the computation of $p_t(r_t)$ involves recursively computing conditional densities for future times. For instance, $p_{t-1}(s_t|r_t)$ for a future t is given by:

$$p_{t-1}(s_t|r_t) = \sum_{r'_t=1,2} p_{t-1}(s_t|r'_t) \cdot p_{t-1}(r'_t|r_t) \tag{23}$$

The multi-period-ahead regime-switching probability $p_{t-1}(r'_t|r_t)$ is also recursive:

$$p_{t-1}(r'_t|r_t) = \sum_{r_{t-1}=1,2} p_{t-1}(r'_t|r_{t-1}) \cdot p_{t-1}(r_{t-1}|r_t) \tag{24}$$

And $p_{t-1}(r_{t-1}|r_t)$ is recursively computed from previous periods:

$$p_{t-1}(r_{t-1}|r_t) = p_{t-2}(r_{t-1}|r_t, s_{t-1}) = \frac{p_{t-2}(s_{t-1}|r_{t-1}) \cdot p_{t-2}(r_{t-1}|r_t)}{\sum_{r'_{t-1}=1,2} p_{t-2}(s_{t-1}|r'_{t-1}) \cdot p_{t-2}(r'_{t-1}|r_t)} \tag{25}$$

3. Empirical Results and Forecasting Performance

The empirical analysis utilized nearly twenty years of daily data (January 3, 1978, to July 23, 1997) for three major U.S. dollar exchange rates: the British pound, German mark, and Japanese yen.

Estimation Results Summary

- Standard GARCH(1,1) models exhibit high volatility persistence, often indicative of parameter instability.
- Regime-switching models with constant regime-specific variances (ARCH(0;0)) effectively identify distinct low and high-volatility regimes, with the high-volatility regime having an unconditional variance 3 to 4 times larger. These regimes are highly persistent, with staying probabilities above 0.975. However, significant conditional heteroskedasticity can remain within regimes.
- Allowing for regime-specific ARCH terms (e.g., ARCH(Q1;4)) reveals more volatility clustering in the high-volatility regime, contrasting with symmetric models.
- The introduction of GARCH terms in the high-volatility regime (RS-GARCH(0,Q1;1,1)) significantly improves the model fit (higher log-likelihood) and removes residual heteroskedasticity compared to regime-switching ARCH, often with fewer parameters. This underscores the importance of both regimes and GARCH effects.
- The proposed RS-GARCH model (Eq. 5) demonstrates a better fit (higher log-likelihood) than Gray's (1996a) variant (Eq. 4), attributed to its more efficient use of conditioning information.
- The improved fit of RS-GARCH over single-regime GARCH is due to lower variance estimates in both tranquil and volatile periods. In volatile periods, this is achieved by smaller estimated direct and subsequent effects of shocks. In tranquil periods, the use of a distinct low-variance regime allows for a quicker reduction in variance estimates, capturing "pressure relieving" shock effects.
- The use of a t-distribution for the error term, instead of a normal distribution, contributes to more stable regimes by better handling outliers.

Forecasting Performance Summary The out-of-sample performance was assessed for one-day ($h = 1$) and ten-day ($h = 10$) horizons using an observed volatility measure $v_{t,h} = \sum_{j=0}^{h-1}(s_{t+j} - \hat{\mu})^2$.

- Forecast Efficiency Regression ($v_{t,h} = \gamma_0 + \gamma_1 V_{t-1}(s_{t,h}) + \eta_t$):

  - For GARCH(1,1) forecasts, the null hypotheses $\gamma_0 = 0$ and $\gamma_1 = 1$ were significantly rejected in eight out of twelve cases, with $\gamma_0 > 0$ and $\gamma_1 < 1$. This implies that high GARCH forecasts generally overestimate the true variance, while low GARCH forecasts do not underestimate it.
  - RS-GARCH models generally show non-rejection of $\gamma_0 = 0$ and $\gamma_1 = 1$, indicating that they successfully reduce the excessive GARCH forecasts.

- Mean Squared Error (MSE): Calculated as the mean of $(v_{t,h} - \hat{V}_{t-1}(s_{t,h}))^2$ over the out-of-sample period.
  - RS-GARCH forecasts exhibited lower MSE than single-regime GARCH in 11 out of 12 cases, with 6 significant improvements.
  - For the one-day horizon, the proposed model showed better MSE than Gray's (1996a) variant in 5 out of 6 cases.

- Restricted R-squared ($\tilde{R}^2$): A measure of forecast quality under the restriction $\gamma_0 = 0$ and $\gamma_1 = 1$:

$$\tilde{R}^2 = 1 - \frac{V[v_{t,h} - \hat{V}_{t-1}(s_{t,h})]}{V[v_{t,h}]} \tag{26}$$

  - $\tilde{R}^2$ values confirmed the outperformance, with RS-GARCH forecasts yielding higher $\tilde{R}^2$ in 11 out of 12 cases compared to single-regime GARCH.
  - The relative outperformance of RS-GARCH over single-regime GARCH was quantified at 22% for the one-day horizon and 58% for the ten-day horizon.
  - Low absolute $\tilde{R}^2$ values are attributed to noise in the observed volatility measure, as shown by Andersen and Bollerslev (1998).

4. Conclusion

The paper concludes that the widely-used GARCH model produces forecasts that are significantly too high in volatile periods due to high persistence of individual shocks. The developed regime-switching GARCH model offers increased flexibility in capturing volatility persistence by employing two distinct volatility regimes with regime-specific GARCH dynamics. Key theoretical advantages include a recursive likelihood function and a convenient recursive procedure for multi-period-ahead volatility forecasting, which improves upon existing RS-GARCH variants. Empirically, the model resolves the issue of excessive single-regime GARCH forecasts in volatile periods and yields significantly better out-of-sample volatility forecasts, with substantial relative improvements in $\tilde{R}^2$ (22% for one-day and 58% for ten-day horizons). The findings underscore the importance of incorporating both regime switches and GARCH effects, allowing for asymmetric heteroskedasticity across regimes, and utilizing a t-distribution for error terms. The paper suggests that further research into the economic mechanisms behind volatility persistence is a promising avenue for improving forecasts.